

2. Assignment - Crypto

1. Encrypt "CHASE YOUR DREAMS" Using

a) Caesar Cipher

Encryption formula - $C = E(P + K) \bmod 26$

where P = numerical value of Plain letter

K = Shift/Key

C = numerical value of ciphertext letter

Decryption formula

$P = D(C - K) \bmod 26$

a) Caesar Cipher (Shift 3) $\therefore K = 3$ (by default)

	P	$(P+K)^3$	Cipher
C	2	5	F
H	7	10	K
A	0	3	D
S	18	21	V
E	4	7	H
Y	24	$27 \xrightarrow{\text{mod}} 1$	B
O	14	17	R
U	20	23	X
R	17	20	U
D	3	6	G
R	17	20	U
E	4	7	H
A	0	3	D
M	12	15	P

b) Shift Cipher with Key 7

→ $K = 7$ (given)

letter(P)	P	$P+K^7$	Cipher
C	2	9	J
H	7	14	O
A	0	7	H
S	18	25	Z
E	4	11	L
Y	24	31 → 5	F
O	14	21	V
U	20	27 → 1	B
R	17	24	Y
D	3	10	K
R	17	24	Y
E	4	11	L
A	0	7	H
M	12	19	T
S	18	25	Z

c) Rot 13

Key (K) = 13

∴ JOH2L EVBY KYLHTF

CHASE

	C	H	A	S	E	Y	O	U	R	D	R	E	A	M	S
P	2	7	0	18	4	24	14	20	17	3	17	4	0	12	18
PH	15	20	B	31	17	37	27	33	30	16	30	17	B	25	31
			↓			↓	↓	↓	↓		↓			↓	
			5			11	1	7	4		4			5	
	P	U	N	F	R	L	B	H	E	Q	E	R	N	Z	F

∴ PUNFRLBHEQERNZF is encrypted text

Decrypt the following

$$P = D(C-14) \bmod 26$$

when value becomes -ve

$$\begin{aligned} P &= (1-3) \bmod 26 \\ &= -2 \bmod 26 \\ &= -2 + 26 = 24 \\ +ve & \text{ — no change} \end{aligned}$$

a) WRWOOB VHFUHW PHVVDJH - Given ciphertext

	$(C-3) \bmod 26$		P
W	$(22-3)$	19	T
R	$(17-3)$	14	O
W	$(22-3)$	19	T
D	$(3-3)$	0	A
O	$(14-3)$	11	L
O	$(14-3)$	11	L
B	$(1-3)$	24	Y
V	$(21-3)$	18	S
H	$(7-3)$	4	E
F	$(5-3)$	2	C
U	$(20-3)$	17	R
H	$(7-3)$	4	E
W	$(22-3)$	19	T
P	$(15-3)$	12	M
H	$(7-3)$	4	E
V	$(21-3)$	18	S
V	$(21-3)$	18	S
D	$(3-3)$	0	A
J	$(9-3)$	6	G
H	$(7-3)$	4	E

∴ decrypted data = TOTALLY SECRET MESSAGE

b) STI 1BTW YNRJ

Shift Cipher S

	C-5	P	
S	18-5	13	N
T	9-5	4	E
I	8-5	3	D
1	8-5	3	D
B	1-5	24 → 22	W
T	19-5	24 → 22	O
W	22-5	17	R
Y	24-5	19	T
N	13-5	08	I
R	17-5	12	M
J	9-5	4	E

Decrypted data:-

∴ NED DWOR Tint

c) FRAQ ZBER DBSSRR

ROT 13

		P
F	$(S+B) \bmod 26 = 18$	S
R	$17+13$	4
A	$0+13$	13
Q	$16+13$	3
Z	$25+13$	12
B	$1+13$	14
E	$4+13$	17
R	$17+13$	4
P	$15+13$	2
S	$1+13$	14
S	$18+13$	5
R	$18+13$	5
R	$17+13$	4
R	$17+13$	4

in ROT13 you ^{can} use

$$P = (K+B) \bmod 26$$

Even for decryption (+ or -)
as $\text{shift} + B \xrightarrow{\text{twice}} \text{gives} +26$

SEND MORE COFFEE